

U.S. Department of Transportation's (USDOT) Open-Source Software Tool To Enable Interoperable Connectivity



The Federal Highway Administration's (FHWA) efforts to enhance interoperable connectivity build on existing open-source software applications that facilitate communication among all parts of a connected deployment, including vehicles, infrastructure, pedestrians, bicyclists, and emergency services. As part of this effort, FHWA developed V2X HubSM,⁽¹⁾ an open-source software application that enables networked, wireless communication between

vehicles, infrastructure, and personal communication devices, creating an interoperable environment for research and development into V2X communications.⁽²⁾

Connected and automated vehicles (CAVs) offer opportunities to improve surface transportation safety and system efficiency. Federal CAV research supports open-source software as a development strategy to provide access and interoperability to users with a variety of hardware and CAV use cases. V2X Hub's development as an open-source software application also allows for a community of practice consisting of users and experts who can support one another by contributing code to address everyday needs and by providing experienced insights.⁽¹⁾ To join the CAV community of practice, email CAVSupportServices@dot.gov.

V2X HUB FUNCTIONALITIES⁽¹⁾

- **Signal phase and timing (SPaT) and MAP messages:** Receives and translates National Transportation Communications for Intelligent Transportation System Protocol (NTCIP) 1202⁽³⁾ objects and combines those objects with preconfigured MAP messages for broadcasting SAE International® (SAE) J2735TM SPaT and MAP messages.^(1,4) V2X Hub also supports the traffic signal controller broadcast message⁽⁴⁾ and J2735 SPaT message verification, as well as Connected Transportation Interoperability (CTI) 4501.⁽⁵⁾
- **Roadside unit (RSU) interface:** Forwards SAE J2735 messages and SAE J3224TM⁽⁶⁾ situational data-sharing messages (SDSMs) from V2X Hub to an RSU for immediate broadcast and listens to incoming SAE J2735 and SAE J3224 messages from an RSU to V2X Hub.⁽⁴⁾ The interface also supports NTCIP 1218⁽⁷⁾ immediate message forward and uses NTCIP 1218 to monitor the health and operational status of RSUs to ensure they are functioning correctly.^(3,4,6)
- **Pedestrian safety:** Uses SAE J3224 SDSMs to share object and vulnerable road user information detected by



Source: USDOT.

infrastructure-based cameras.⁽⁶⁾ The RSU packages and broadcasts this information to nearby CAVs, enhancing situational awareness and safety for all road users.

- **CARMA CloudSM (traffic control, work zone):** Establishes a network connection with a server running CARMA Cloud, enabling work zone and emergency response vehicle use cases.⁽⁸⁾



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- **CARMA StreetsSM**: Feeds CARMA Streets (an infrastructure edge-computing component) with real-time data for executing transportation use cases.⁽⁹⁾ Currently, CARMA Streets enables signal optimization and cooperative perception use cases, leveraging SAE J3224⁽⁶⁾ SDSMs to enhance awareness and coordination between infrastructure and connected vehicles.⁽⁹⁾
- **CDASim**: Establishes a bridge between V2X Hub⁽¹⁾ and the CDASim cosimulation platform⁽¹⁰⁾, enabling V2X Hub to seamlessly interact with other platforms, including CARMA Platform^{SM(11)}, CARLA^{®(12)}, Simulation of Urban MObility (SUMOTM)⁽¹³⁾, and ns3⁽¹⁴⁾ within the simulation environment.
- **CAV Telematics Tool**: Feeds real-time data into the telematics tool, enabling the creation of customized dashboards for visualizing message data. Telematics dashboards allow real-time monitoring and analysis of cooperative driving automation use case testing, including vehicle and traffic data.⁽¹⁵⁾
- **Traveler information message (TIM)**: Generates and broadcasts SAE J2735⁽⁴⁾ TIMs containing roadway or



Source: FHWA. (See references 1, 8–10, and 15.)

operational updates (e.g., work zones, incidents, or speed advisories) to connected vehicles via RSUs. The TIM plug-in enhances situational awareness and supports safe and efficient traveler communication between infrastructure and road users.

- **Operational data environment**: Pushes data to an operational data environment server, where the data can be distributed to devices and transportation management applications.⁽²⁾

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